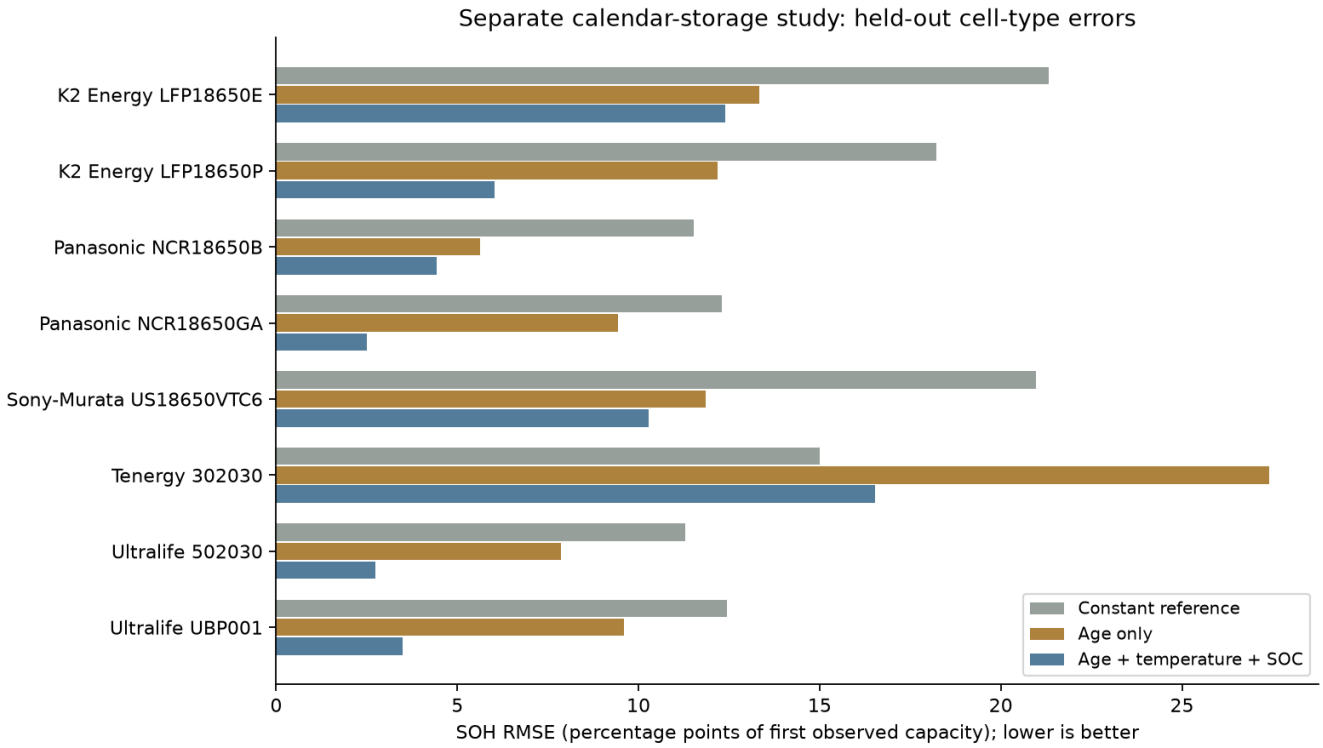
ReCell | Calendar-aging research review

27 October 2026 | Independent study | Protocol calendar_validation_v1

Review decision: retain this as a research comparison. Average prediction error improved, but severe failures prevent adoption as a battery-life or dispatch model.

Held-out-type RMSE	Age only	Age + temperature + SOC
Percentage points	12.16	7.29

The primary test covers **218 physical cells, eight held-out cell types and 9,249 future capacity observations**. Types receive equal weight in the final score.



218 cells / 8 types. C2 has 44 negative predictions; no M50T calibration.

Failure evidence: 44 impossible negative SOH predictions across 4 cells; worst absolute error 73.71 percentage points. Every failure remains in the reported metrics.

This is separate from the first-life transfer experiment. No result here establishes M50T calibration, Indian pack performance, measured life extension or customer savings.

1. Data and attributes

The pinned Stanford calendar release supplies storage experiments across eight base cell types and four chemistries. It contains Sony VTC6 NMC cells, but no LG M50T cells.

Attribute	Role in this study
Third C/5 discharge capacity (Ah)	RPT0.2C_2_D_capacity supplies the measured target and each cell's first reference.
Elapsed storage days	Predictor in C1 and C2; clock/reference consistency is audited.
Storage temperature (degrees C)	C2 predictor; source settings: 24, 45, 60 or 85.
Storage SOC (fraction)	C2 predictor; source settings: 0.5 or 1.0.
Physical-cell identity and base type	Split and weighting keys, not predictive inputs.
Chemistry, lot and author comments	Compatibility review and predefined sensitivity; alternate _v2 lots stay grouped with their base type.

Coverage: 259 inventory IDs; 232 published summary files; 27 missing summaries. The parser accepts 9,481 checkpoints and quarantines 14 ambiguous day-only clock rows across seven cells. Fourteen single-observation cells remain descriptive; they do not receive invented future labels. The other 218 support predictive evaluation.

What is fitted, and what is held back?

Each of eight outer folds holds out one complete base cell type for testing. The next sorted type is reserved for validation; the other six supply parameter fitting. No physical battery appears in multiple roles within a fold. Test cells provide only their initial observed capacity reference, with no parameter updates.

Primary cohort: all eligible author-curated histories. **Sensitivity:** remove all 13 cells with notes about deleted potentially swapped measurements from every role. Recoveries, above-reference capacities and double-bag notes remain visible.

Summary capacities are source-provided labels. All underlying raw diagnostic curves were not independently audited. This limitation applies to every model.

2. Mathematical model and fitting

For cell c, Q(c,0) is the first accepted measured capacity and Q(c,i) is a later capacity. Elapsed age d is measured from that first accepted reference. SOH is a fraction of this reference, not manufacturer nominal capacity.

```
y(c,i) = Q(c,i) / Q(c,0)
u = sqrt(d / 365.25)
zT = 1000 * [1/297.15 - 1/(T_C + 273.15)]
zS = s - 0.5
```

Model	Predicted SOH	Interpretation
C0	1	Constant initial-reference baseline
C1	1 - a*u, with a >= 0	Shared square-root-age loss
C2	1 - exp(alpha + beta_T*zT + beta_s*zS)*u	Age loss conditioned on storage temperature and SOC

The temperature transform is centered at 24 degrees C (297.15 K); SOC is centered at 50%. C2 is a fixed empirical model family, not an identified electrochemical law. Its exponential coefficient is positive, but predictions can fall below zero.

Training objective

```
minimize sum_(k,c,i) w(k,c,i) * [yhat(c,i) - y(c,i)]^2
w(k,c,i) = 1 / [K * C_k * N_c]
```

K is the number of represented training types, C_k their training-cell count and N_c a cell's permitted positive-age observations. Thus each type, then cell, receives equal weight. The reference observation at age zero is excluded from fitting and scoring.

```
C1: a = max(0, sum(w*u*(1-y)) / sum(w*u*u))
```

C2 uses bounded nonlinear least squares: alpha in [-20, 2], beta_T in [0, 20], beta_s in [-10, 10]. Six fixed starts combine beta_T = 0, 2, 5 with beta_s = 0, 1; initial alpha is log(max(a, 1e-8)), restricted to optimizer bounds. The converged solution with lowest training cost wins. Validation/test labels never select it.

SciPy trf, linear loss, at most 2,000 evaluations per start, x_scale=1, ftol=xtol=gtol=1e-8. No label or prediction clipping; no interpolation.

3. Metrics, uncertainty and failures

```
RMSE_c = sqrt(mean_i[(yhat(c,i) - y(c,i))^2])
RMSE_type = mean_cells(RMSE_c)
Reported RMSE = mean_8_test_types(RMSE_type)
```

Multiply a SOH-fraction error by 100 to obtain percentage points. A one-point error equals 1% of that cell's first measured capacity. MAE follows the same cell/type aggregation using absolute errors. Lower error is better.

Held-out base type	C0	C1	C2
K2 Energy LFP18650E	21.33	13.33	12.39
K2 Energy LFP18650P	18.23	12.18	6.02
Panasonic NCR18650B	11.52	5.63	4.43
Panasonic NCR18650GA	12.31	9.44	2.51
Sony-Murata US18650VTC6	20.98	11.86	10.27
Tenergy 302030	15.00	27.40	16.52
Ultralife 502030	11.30	7.86	2.73
Ultralife UBP001	12.43	9.60	3.48

Primary contrast: C2 minus C1 = -4.87 points; descriptive 95% bootstrap interval [-7.17, -2.73]. Ten thousand paired resamples use the eight types as units, seed 42. Overlapping training folds and eight types limit the inference; the observations are not thousands of independent battery types.

Registered secondary check	C1 RMSE	C2 RMSE
Train age <=180d, test >180d	17.11	10.65
Exclude swap-note cells, full age	11.79	7.16

Do not hide failures: the 44 negative C2 outputs occur on four K2 Energy LFP18650E cells at 85 degrees C / 100% SOC. Tenergy's C2 error exceeds the constant baseline; its worst single absolute error is 73.71 points. No source-data explanation is assumed without further raw evidence.

4. What this supports, and what remains open

Supported: a reproducible comparison of three preregistered apparent storage-loss models on unseen cell types, including chronological extrapolation and a predefined author-note sensitivity. The conditioned model has lower average error.

Model limit: severe type errors and physically impossible forecasts prevent automatic adoption. Any changed family needs a new registered comparison, not retuning on these test results.

Aging accounting: storage experiments include periodic diagnostic cycling. Pure calendar loss and diagnostic loss are not separately identified. Adding this curve to existing mixed-aging wear would risk double-counting.

Application gap: M50T cells are absent. Measured target-cell temperature, independent calibration and pack/BMS evidence are still needed. Ambient weather is not cell temperature. This study does not validate Indian customer savings.

Source gate: first-life transfer remains blocked by 13 rejected cycling batches. This calendar study neither repairs that data nor demonstrates positive or negative transfer.

Release gap: the reviewed OSF node metadata declares no license. This records a missing declaration; intended redistribution/commercial use still needs clarification.

Sources and reproducibility

Author source: Joule-Decade-Calendar-Aging, commit 0bc2d6393da223e6d850896b6501153630d393ca; OSF ju325 release. Third-discharge summaries are pinned by archive and member hashes.

Sealed before fitting: 2026-10-27T17:53:41.066944+00:00. Protocol SHA-256:

```
7377d66d0867b7500e082ca61382dc80
7a9d11537e85354cb4223355661c5668
```

Review evidence: docs/calendar_source_audit.md; configs/calendar_protocol_v1.yaml; outputs/calendar/calendar_validation_v1/. The validation folder retains fit coefficients, all predictions, cell/type metrics, optimizer records, implementation source and input/artifact hashes.

Verification: 141 software tests and lint pass. An independent saved-output check reproduced all scores and predictions across 32 fold configurations, verified physical/type separation, and checked 200,628 saved prediction rows and artifact hashes. Software checks do not establish field validity.

Historical forecasting, dispatch and dashboard results remain preserved. This PDF is an additional review artifact; no model was refitted to produce it.